

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Pharm (Pharmaceutical Technology)

University Theory Examination May 2015

PH819.08 Novel Drug Delivery Systems - II

Date: 13/05/2015 , Wednesday

Time: 10:00 a.m to 01:00 p.m

Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

- Q.1** **A** Write note on factors affecting nasal drug absorption. 05
- B** What is drug targeting. What are the levels of drug targeting. Differentiate between active & passive drug targeting. 05
- Q.2** Enlist the approaches to colon-specific drug delivery. Explain any one in detail. 10

OR

- Q.2** **A** Write note on mechanism of drug transport across nasal mucosa. 05
- B** Enlist novel ocular drug delivery systems. Explain in detail "Implants". 05
- Q.3** Answer the following (Any three) 15
- A** Write note on applications of intranasal drug delivery. 05
- B** Write note on physiological characteristics of the colon. 05
- C** Explain barriers to restrict intraocular drug transport. 05
- D** Using suitable example, explain rationale for targeted drug delivery system. 05

SECTION-II

- Q.4** Discuss in detail the particle deposition mechanism in respiratory tract; emphasizing on size and velocities of particles. 10
- Q.5** Enumerate the approaches for drug delivery to brain. Discuss inwardly directed transport systems in detail. 10

OR

- Q.5** What are the barriers in targeting drugs to tumors? What are the approaches to overcome these barriers? 10
- Q.6** Answer the following (Any three) 15
- A** Write a short note on pMDI. 05
- B** Write a note on in vitro measurement of brain uptake. 05
- C** How pathology of cancer affects formulation approaches for drug targeting to tumors? 05
- D** Write a short note on invitro evaluation of particle deposition efficacy of inhaleables. 05